

Chapter three

Limits and Continuity

Definition: Let f be a function defined at each point of some open interval containing a , except possibly at a itself. Then a number L is the limit of $f(x)$ as x approaches a (or is the limit of f at a) if for every $\varepsilon > 0$ there is a number $\delta > 0$ such that if $0 < |x - a| < \delta$, then $|f(x) - L| < \varepsilon$.

If L is the limit of $f(x)$ as x approaches a , then we write

$$\lim_{x \rightarrow a} f(x) = L$$

Example

prove that $\lim_{x \rightarrow 3} 3x - 5 = 5$

Solution :(i) preliminary analysis of the problem (guessing a value for δ)

Let $\varepsilon > 0$ be given. We want to find a number $\delta > 0$ such that

$$|3x - 4 - 5| < \varepsilon \text{ whenever } 0 < |x - 3| < \delta. \text{ But } |3x - 4 - 5| = |3x - 9| = 3|x - 3|$$

Therefore, we want $3|x - 3| < \varepsilon$ whenever $0 < |x - 3| < \delta$.

That means $|x - 3| < \frac{\varepsilon}{3}$ whenever $0 < |x - 3| < \delta$

This suggests that we should choose $\delta = \frac{\varepsilon}{3}$

Example: Show that $\lim_{x \rightarrow 2} \frac{x^2}{x+2} = 2$

Proof: Let $\varepsilon > 0$ be given, we need to find $\delta > 0$ such that

$$0 < |x-2| < \delta \Rightarrow \left| \frac{x^2}{x+2} - 2 \right| < \varepsilon \Rightarrow \frac{|x+1||x-2|}{|x+2|} < \varepsilon$$

Assume $\delta \leq 1$. Then $|x+1| < 4$. Since $|x-2| < 1 \Rightarrow -1 < x-2 < 1$

$$\Rightarrow 3 < x+2 < 5 \Rightarrow \frac{1}{5} < \frac{1}{x+2} < \frac{1}{3} \Rightarrow \frac{1}{|x+2|} < \frac{1}{3}$$

$$\Rightarrow \frac{|x+1||x-2|}{|x+2|} < \varepsilon \Rightarrow \frac{4}{3}|x-2| < \varepsilon$$

$$\Rightarrow |x-2| < \frac{3}{4}\varepsilon$$

Now, choose $\delta = \min(1, \frac{3}{4}\varepsilon)$. Therefore $\lim_{x \rightarrow 2} \frac{x^2}{x+2} = 2$.

Example: Prove that $\lim_{x \rightarrow 3} \sqrt{x+1} = 2$

Proof: Let $\varepsilon > 0$ be given, we need to find $\delta > 0$ such that

$$0 < |x-3| < \delta \Rightarrow |\sqrt{x+1} - 2| < \varepsilon$$

$$\text{But } |\sqrt{x+1} - 2| = \frac{|x-3|}{\sqrt{x+1} + 2} < \varepsilon. \text{ Assume } \delta \leq 1. \text{ Then}$$

$$-1 < x-3 < 1 \Rightarrow 3 < x+1 < 5 \Rightarrow \sqrt{3} < \sqrt{x+1}$$

$$\Rightarrow \sqrt{3} + 2 < \sqrt{x+1} + 2 \Rightarrow \frac{1}{\sqrt{x+1} + 2} < \frac{1}{\sqrt{3} + 2}$$

$$\Rightarrow |x-3| = \frac{|x-3|}{\sqrt{x+1} + 2} < \frac{|x-3|}{\sqrt{3} + 2} < \varepsilon$$

$$\text{Now choose } \delta = \min(1, (\sqrt{3} + 2)\varepsilon)$$

$$\text{Therefore } \lim_{x \rightarrow 3} \sqrt{x+1} = 2.$$

Exercises: Prove the following using $(\varepsilon - \delta)$ definition

$$1. \lim_{x \rightarrow 3} \frac{x+3}{x-1} = 3 \quad 2. \lim_{x \rightarrow \frac{-1}{2}} \frac{x}{x+1} = -1 \quad 3. \lim_{x \rightarrow -1} \frac{x^2+2}{2x-1} = -1 \quad 4. \lim_{x \rightarrow \frac{1}{2}} \frac{1}{x} = 2$$

$$5. \lim_{x \rightarrow 0} \frac{x}{x^2+1} = 0. \quad 6. \lim_{x \rightarrow -1} x^2 + 6x + 5 = 12.$$

Theorem 1 (Uniqueness of Limits)

If the limit of a function f at a exists, then this limit is unique. Equivalently, if L and M are both limits of f at a , then $L = M$

Basic Limit Theorems

Theorem 2 (Sum Theorem for Limits)

Let $\lim_{x \rightarrow a} f(x)$ and $\lim_{x \rightarrow a} g(x)$ exists. Then $\lim_{x \rightarrow a} [f(x) + g(x)]$ exists , and

$$\lim_{x \rightarrow a} [f(x) + g(x)] = \lim_{x \rightarrow a} f(x) + \lim_{x \rightarrow a} g(x)$$

Theorem 3 (Constant Multiple Theorem for Limits)

Let $\lim_{x \rightarrow a} f(x)$ exists, then $\lim_{x \rightarrow a} Kf(x)$ exists and

$$\lim_{x \rightarrow a} Kf(x) = K \lim_{x \rightarrow a} f(x) \text{ for any real number } K$$

Theorem 4 (Product Theorem for Limits)

Let $\lim_{x \rightarrow a} f(x)$ and $\lim_{x \rightarrow a} g(x)$ exists. Then $\lim_{x \rightarrow a} f(x).g(x)$ exists, and

$$\lim_{x \rightarrow a} f(x).g(x) = \lim_{x \rightarrow a} f(x). \lim_{x \rightarrow a} g(x)$$

Theorem 5 (Quotient theorem for limits)

Let $\lim_{x \rightarrow a} f(x)$ and $\lim_{x \rightarrow a} g(x)$ exists. If $\lim_{x \rightarrow a} g(x) \neq 0$,

$$\text{then } \lim_{x \rightarrow a} \frac{f(x)}{g(x)} \text{ exists and } \lim_{x \rightarrow a} \frac{f(x)}{g(x)} = \frac{\lim_{x \rightarrow a} f(x)}{\lim_{x \rightarrow a} g(x)}$$

Let us proof Sum Theorem for Limits and the others can be proved similarly.

Let $\lim_{x \rightarrow a} f(x) = L$ and $\lim_{x \rightarrow a} g(x) = M$

WTS: $\lim_{x \rightarrow a} [f(x) + g(x)] = L + M$

i.e. Given $\varepsilon > 0$, $\exists \delta > 0 \ni 0 < |x - a| < \delta \Rightarrow |[f(x) + g(x)] - [L + M]| < \varepsilon$

Let $\varepsilon > 0$. Since $\lim_{x \rightarrow a} f(x) = L \exists \delta_1 > 0 \ni 0 < |x - a| < \delta_1 \Rightarrow |f(x) - L| < \frac{\varepsilon}{2}$ and since

$$\lim_{x \rightarrow a} g(x) = M \exists \delta_2 > 0 \ni 0 < |x - a| < \delta_2 \Rightarrow |g(x) - M| < \frac{\varepsilon}{2}$$

Take $\delta = \min\{\delta_1, \delta_2\}$

$$\Rightarrow 0 < |x - a| < \delta$$

$$\text{Now } |[f(x) + g(x)] - [L + M]|$$

$$= |f(x) - L + g(x) - M|$$

$$\leq |f(x) - L| + |g(x) - M| \quad (\text{triangle inequality})$$

$$< \frac{\varepsilon}{2} + \frac{\varepsilon}{2} = \varepsilon. \text{ This completes the proof}$$

To verify this, consider $f(x) = 5x^2 + 2x + 3$

and $g(x) = x + 2$.

Then
$$\lim_{x \rightarrow 0} f(x) = \lim_{x \rightarrow 0} (5x^2 + 2x + 3)$$

$$= 5 \lim_{x \rightarrow 0} x^2 + 2 \lim_{x \rightarrow 0} x + 3 = 3$$

$$\lim_{x \rightarrow 0} g(x) = \lim_{x \rightarrow 0} (x + 2) = \lim_{x \rightarrow 0} x + 2 = 2$$

$$\therefore \lim_{x \rightarrow 0} (5x^2 + 2x + 3) \lim_{x \rightarrow 0} (x + 2) = 6 \quad \text{.....(i)}$$

Again $\lim_{x \rightarrow 0} [f(x) \cdot g(x)] = \lim_{x \rightarrow 0} [(5x^2 + 2x + 3)(x + 2)]$

$$= \lim_{x \rightarrow 0} (5x^3 + 12x^2 + 7x + 6)$$

$$= 5 \lim_{x \rightarrow 0} x^3 + 12 \lim_{x \rightarrow 0} x^2 + 7 \lim_{x \rightarrow 0} x + 6$$

$$= 6 \quad \text{.....(ii)}$$

From (i) and (ii), $\lim_{x \rightarrow 0} [(5x^2 + 2x + 3)(x + 2)] = \lim_{x \rightarrow 0} (5x^2 + 2x + 3) \lim_{x \rightarrow 0} (x + 2)$

$$4. \lim_{x \rightarrow a} \left\{ \frac{f(x)}{g(x)} \right\} = \frac{\lim_{x \rightarrow a} f(x)}{\lim_{x \rightarrow a} g(x)} \quad \text{provided } \lim_{x \rightarrow a} g(x) \neq 0$$

To verify this, consider the function $f(x) = \frac{x^2 + 5x + 6}{x + 2}$

$$\begin{aligned} \text{we have } \lim_{x \rightarrow -1} (x^2 + 5x + 6) &= (-1)^2 + 5(-1) + 6 \\ &= 1 - 5 + 6 \\ &= 2 \end{aligned}$$

$$\begin{aligned} \text{and } \lim_{x \rightarrow -1} (x + 2) &= -1 + 2 \\ &= 1 \end{aligned}$$

$$\frac{\lim_{x \rightarrow -1} (x^2 + 5x + 6)}{\lim_{x \rightarrow -1} (x + 2)} = \frac{2}{1} = 2$$

Example
evaluate the following limits

Evaluate : $\lim_{x \rightarrow 2} \frac{\sqrt{3-x} - 1}{2-x}$.

Solution : Rationalizing the numerator, we have

$$\frac{\sqrt{3-x}-1}{2-x} = \frac{\sqrt{3-x}-1}{2-x} \times \frac{\sqrt{3-x}+1}{\sqrt{3-x}+1}$$

$$= \frac{3-x-1}{(2-x)(\sqrt{3-x}+1)}$$

$$= \frac{2-x}{(2-x)(\sqrt{3-x}+1)}$$

$$\therefore \lim_{x \rightarrow 2} \frac{\sqrt{3-x}-1}{2-x} = \lim_{x \rightarrow 2} \frac{2-x}{(2-x)(\sqrt{3-x}+1)}$$

$$= \lim_{x \rightarrow 2} \frac{1}{(\sqrt{3-x}+1)}$$

$[\because x \neq 2]$

$$= \frac{1}{(\sqrt{3-2}+1)}$$

$$= \frac{1}{1+1} = \frac{1}{2}$$

$$\text{Evaluate : } \lim_{x \rightarrow 3} \frac{\sqrt{12-x} - x}{\sqrt{6+x} - 3}.$$

Solution : Rationalizing the numerator as well as the denominator, we get

$$\begin{aligned} \lim_{x \rightarrow 3} \frac{\sqrt{12-x} - x}{\sqrt{6+x} - 3} &= \lim_{x \rightarrow 3} \frac{(\sqrt{12-x} - x)(\sqrt{12-x} + x)(\sqrt{6+x} + 3)}{\sqrt{6+x} - 3(\sqrt{6+x} + 3)(\sqrt{12-x} + x)} \\ &= \lim_{x \rightarrow 3} \frac{(12 - x - x^2)}{6 + x - 9} \cdot \lim_{x \rightarrow 3} \frac{\sqrt{6+x} + 3}{\sqrt{12-x} + x} \\ &= \lim_{x \rightarrow 3} \frac{-(x+4)(x-3)}{(x-3)} \cdot \lim_{x \rightarrow 3} \frac{\sqrt{6+x} + 3}{\sqrt{12-x} + x} \quad [\because x \neq 3] \\ &= -(3+4) \frac{6}{6} = -7 \end{aligned}$$

Evaluate : $\lim_{x \rightarrow 2} \frac{x^3 - 8}{x - 2}$.

Solution : $\lim_{x \rightarrow 2} \frac{x^3 - 8}{x - 2}$

$$= \lim_{x \rightarrow 2} \frac{(x - 2)(x^2 + 2x + 4)}{(x - 2)}$$

$$= \lim_{x \rightarrow 2} (x^2 + 2x + 4) \quad [\because x \neq 2]$$

$$= 2^2 + 2 \times 2 + 4$$

$$= 12$$

Theorem 6 (Substitution Theorem for Limits)

Suppose $\lim_{x \rightarrow a} f(x) = c$, and $f(x) \neq c$ for all x in some open interval about a , with the

possible exception of a itself. Suppose also that $\lim_{y \rightarrow c} g(y)$ exists. Then

$$\lim_{x \rightarrow a} g(f(x)) = \lim_{y \rightarrow c} g(y)$$

Theorem 7 (Squeezing Theorem)

Assume that $f(x) \leq g(x) \leq h(x)$ for all x in some open interval about a , except possibly at

a itself. If $\lim_{x \rightarrow a} f(x) = L$ and $\lim_{x \rightarrow a} h(x) = L$, then

$$\lim_{x \rightarrow a} g(x) \text{ exists and } \lim_{x \rightarrow a} g(x) = L$$

Prove that $\lim_{x \rightarrow 0} \frac{\sin x}{x} = 1$

Proof : Draw a circle of radius 1 unit and with centre at the origin O . Let $B (1,0)$ be a point on the circle. Let A be any other point on the circle. Draw $AC \perp OX$.

Let $\angle AOX = x$ radians, where $0 < x < \frac{\pi}{2}$

Draw a tangent to the circle at B meeting OA produced at D . Then $BD \perp OX$.

Area of $\triangle AOC <$ area of sector $OBA <$ area of $\triangle OBD$.

$$\text{or } \frac{1}{2} OC \times AC < \frac{1}{2} x(1)^2 < \frac{1}{2} OB \times BD$$

$$\left[\because \text{area of triangle} = \frac{1}{2} \text{base} \times \text{height and area of sector} = \frac{1}{2} \theta r^2 \right]$$

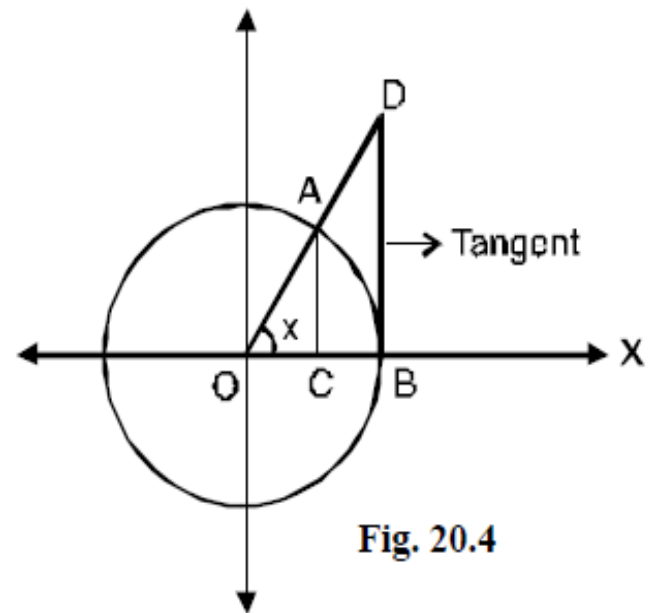


Fig. 20.4

$$\therefore \frac{1}{2} \cos x \sin x < \frac{1}{2} x < \frac{1}{2} \cdot 1 \cdot \tan x$$

$$\left[\because \cos x = \frac{OC}{OA}, \sin x = \frac{AC}{OA} \text{ and } \tan x = \frac{BD}{OB}, OA = 1 = OB \right]$$

$$\text{i.e., } \cos x < \frac{x}{\sin x} < \frac{\tan x}{\sin x} \quad \left[\text{Dividing throughout by } \frac{1}{2} \sin x \right]$$

$$\text{or } \cos x < \frac{x}{\sin x} < \frac{1}{\cos x}$$

$$\text{or } \frac{1}{\cos x} > \frac{\sin x}{x} < \cos x$$

$$\text{i.e., } \cos x < \frac{\sin x}{x} < \frac{1}{\cos x}$$

Taking limit as $x \rightarrow 0$, we get

$$\lim_{x \rightarrow 0} \cos x < \lim_{x \rightarrow 0} \frac{\sin x}{x} < \lim_{x \rightarrow 0} \frac{1}{\cos x}$$

or $1 < \lim_{x \rightarrow 0} \frac{\sin x}{x} < 1$

$$\left[\because \lim_{x \rightarrow 0} \cos x = 1 \text{ and } \lim_{x \rightarrow 0} \frac{1}{\cos x} = \frac{1}{1} = 1 \right]$$

Thus, $\lim_{x \rightarrow 0} \frac{\sin x}{x} = 1$

Evaluate $\lim_{x \rightarrow 0} \frac{1 - \cos x}{2x^2}$.

Solution : $\lim_{x \rightarrow 0} \frac{1 - \cos x}{2x^2} = \lim_{x \rightarrow 0} \frac{2 \sin^2 \frac{x}{2}}{2x^2}$ $\left[\begin{array}{l} \because \cos 2x = 1 - 2 \sin^2 x, \\ \therefore 1 - \cos 2x = 2 \sin^2 x \\ \text{or } 1 - \cos x = 2 \sin^2 \frac{x}{2} \end{array} \right]$

$$= \lim_{x \rightarrow 0} \left(\frac{\sin \frac{x}{2}}{2 \times \frac{x}{2}} \right)^2 \quad [\text{Multiplying and dividing the denominator by 2}]$$

$$= \frac{1}{4} \lim_{\frac{x}{2} \rightarrow 0} \left(\frac{\sin \frac{x}{2}}{\frac{x}{2}} \right)^2$$

$$= \frac{1}{4} \times 1 = \frac{1}{4}$$

$$\therefore \lim_{x \rightarrow 0} \frac{1 - \cos x}{2x^2} = \frac{1}{4}$$

Evaluate : $\lim_{\theta \rightarrow 0} \frac{1 - \cos 4\theta}{1 - \cos 6\theta}$

Solution : $\lim_{\theta \rightarrow 0} \frac{1 - \cos 4\theta}{1 - \cos 6\theta} = \lim_{\theta \rightarrow 0} \frac{2 \sin^2 2\theta}{2 \sin^2 3\theta}$

$$= \lim_{\theta \rightarrow 0} \left(\left(\frac{\sin 2\theta}{2\theta} \times 2\theta \right)^2 \left(\frac{3\theta}{\sin 3\theta} \times \frac{1}{3\theta} \right)^2 \right)$$

$$= \lim_{\theta \rightarrow 0} \left(\frac{\sin 2\theta}{2\theta} \right)^2 \left(\frac{3\theta}{\sin 3\theta} \right) \frac{4\theta^2}{9\theta^2}$$

$$= \left(\frac{4}{9} \right) \lim_{2\theta \rightarrow 0} \left(\frac{\sin 2\theta}{2\theta} \right)^2 \lim_{3\theta \rightarrow 0} \left(\frac{3\theta}{\sin 3\theta} \right)$$

$$= \frac{4}{9} \times 1 \times 1 = \frac{4}{9}$$

Exercises

1) Evaluate each of the following:

$$(a) \lim_{x \rightarrow 0} \frac{\sqrt{4+x} - \sqrt{4-x}}{x}$$

$$(b) \lim_{x \rightarrow 0} \frac{\sqrt{2+x} - \sqrt{2}}{x}$$

$$(c) \lim_{x \rightarrow 3} \frac{\sqrt{3+x} - \sqrt{6}}{x-3}$$

$$(d) \lim_{x \rightarrow 0} \frac{x}{\sqrt{1+x} - 1}$$

$$(e) \lim_{x \rightarrow 2} \frac{\sqrt{3x-2} - x}{2 - \sqrt{6-x}}$$

2. Evaluate each of the following:

$$(a) \lim_{x \rightarrow 0} \frac{1 - \cos x}{x^2}$$

$$(b) \lim_{x \rightarrow 0} \frac{1 - \cos 8x}{x}$$

$$(c) \lim_{x \rightarrow 0} \frac{\sin 2x(1 - \cos 2x)}{x^3}$$

$$(d) \lim_{x \rightarrow 0} \frac{1 - \cos 2x}{3 \tan^2 x}$$

3. Evaluate the following :

$$(a) \lim_{x \rightarrow \pi} \frac{\sin x}{\pi - x}$$

$$(b) \lim_{x \rightarrow 1} \frac{\cos \frac{\pi}{2} x}{1 - x}$$

$$(c) \lim_{x \rightarrow \frac{\pi}{2}} (\sec x - \tan x)$$

One sided limits

Definitions: i) Let f be a function defined in some interval (a, b) . A number L is the limit of $f(x)$ as x approaches a from the right if $\forall \varepsilon > 0, \exists \delta > 0$ such that $0 < x - a < \delta \Rightarrow |f(x) - L| < \varepsilon$. In this case we write $\lim_{x \rightarrow a^+} f(x) = L$.

ii) A number L is called a left hand limit of f if

$\forall \varepsilon > 0, \exists \delta > 0$ such that $-\delta < x - a < 0 \Rightarrow |f(x) - L| < \varepsilon$. In this case we write $\lim_{x \rightarrow a^-} f(x) = L$.

Examples: Using ($\varepsilon - \delta$) definitions show that

$$\text{a) } \lim_{x \rightarrow \frac{5}{6}^-} \sqrt{5-6x} = 0 \quad \text{b) } \lim_{x \rightarrow \frac{1}{2}^+} \sqrt{2x-1} = 0 \quad \text{c) } \lim_{x \rightarrow 9^-} \sqrt[4]{9-x} = 0$$

Solution: a) Let $\varepsilon > 0$ be given, we want to find a number $\delta > 0$ such that

$$-\delta < x - \frac{5}{6} < 0 \Rightarrow |\sqrt{5-6x}| < \varepsilon. \text{ But } |\sqrt{5-6x}| = \sqrt{5-6x} < \varepsilon$$

$$\Rightarrow 5-6x < \varepsilon^2 \Rightarrow \frac{5}{6} - x < \frac{\varepsilon^2}{6} \Rightarrow x - \frac{5}{6} > -\frac{\varepsilon^2}{6}$$

$$\Rightarrow \frac{-\varepsilon^2}{6} < x - \frac{5}{6}.$$

Now, choose $\delta = \frac{\varepsilon^2}{6}$. Therefore $\lim_{x \rightarrow \frac{5}{6}^-} \sqrt{5-6x} = 0$

Theorem: Let f be defined on an open interval about a , except possibly at a itself. Then

$\lim_{x \rightarrow a} f(x)$ exists if and only if both one sided limits $\lim_{x \rightarrow a^+} f(x)$ and $\lim_{x \rightarrow a^-} f(x)$ exist and

$\lim_{x \rightarrow a^+} f(x) = \lim_{x \rightarrow a^-} f(x)$. In that case $\lim_{x \rightarrow a} f(x) = \lim_{x \rightarrow a^+} f(x) = \lim_{x \rightarrow a^-} f(x)$

Example: 1) Let $f(x) = \begin{cases} 3x+1 & \text{for } x < 1 \\ x+3 & \text{for } x > 1 \end{cases}$. Find $\lim_{x \rightarrow 1} f(x)$

Solution: $\lim_{x \rightarrow 1^+} f(x) = \lim_{x \rightarrow 1^+} x+3 = 4$ and $\lim_{x \rightarrow 1^-} f(x) = \lim_{x \rightarrow 1^-} 3x+1 = 4$. By the above theorem, we have

$\lim_{x \rightarrow 1} f(x) = 4$.

2) Prove that $\lim_{x \rightarrow 0} \frac{|x|}{x}$ does not exist.

Find the value of 'a' so that

$$\lim_{x \rightarrow 1} f(x) \text{ exist, where } f(x) = \begin{cases} 3x + 5, & x \leq 1 \\ 2x + a, & x > 1 \end{cases}$$

Solution : $\lim_{x \rightarrow 1^-} f(x) = \lim_{x \rightarrow 1} (3x + 5) \quad [\because f(x) = 3x + 5 \text{ for } x \leq 1]$

$$= \lim_{h \rightarrow 0} [3(1-h) + 5]$$

$$= 3 + 5 = 8 \quad \dots(i)$$

$$\lim_{x \rightarrow 1^+} f(x) = \lim_{x \rightarrow 1} (2x + a) \quad [\because f(x) = 2x + a \text{ for } x > 1]$$

$$= \lim_{h \rightarrow 0} (2(1+h) + a)$$

$$= 2 + a \quad \dots(ii)$$

We are given that $\lim_{x \rightarrow 1} f(x)$ will exist provided

$$\Rightarrow \lim_{x \rightarrow 1^-} f(x) = \lim_{x \rightarrow 1^+} f(x)$$

$\therefore$ From (i) and (ii),

$$2 + a = 8$$

$$\therefore \text{or, } a = 6$$

1. Evaluate each of the following limits :

$$(a) \lim_{x \rightarrow 2} [2(x+3) + 7]$$

$$(b) \lim_{x \rightarrow 0} (x^2 + 3x + 7)$$

$$(c) \lim_{x \rightarrow 1} [(x+3)^2 - 16]$$

$$(d) \lim_{x \rightarrow 1} [(x+1)^2 + 2]$$

$$(e) \lim_{x \rightarrow 0} [(2x+1)^3 - 5]$$

$$(f) \lim_{x \rightarrow 1} (3x+1)(x+1)$$

2. Find the limits of each of the following functions :

$$(a) \lim_{x \rightarrow 5} \frac{x-5}{x+2}$$

$$(b) \lim_{x \rightarrow 1} \frac{x+2}{x+1}$$

$$(c) \lim_{x \rightarrow -1} \frac{3x+5}{x-10}$$

$$(d) \lim_{x \rightarrow 0} \frac{px+q}{ax+b}$$

$$(e) \lim_{x \rightarrow 3} \frac{x^2-9}{x-3}$$

$$(f) \lim_{x \rightarrow 5} \frac{x^2-25}{x+5}$$

$$(g) \lim_{x \rightarrow 2} \frac{x^2 - x - 2}{x^2 - 3x + 2}$$

$$(h) \lim_{x \rightarrow \frac{1}{3}} \frac{9x^2 - 1}{3x - 1}$$

3. Evaluate each of the following limits:

$$(a) \lim_{x \rightarrow 1} \frac{x^3 - 1}{x - 1}$$

$$(b) \lim_{x \rightarrow 0} \frac{x^3 + 7x}{x^2 + 2x}$$

$$(c) \lim_{x \rightarrow 1} \frac{x^4 - 1}{x - 1}$$

$$(d) \lim_{x \rightarrow 1} \left[\frac{1}{x-1} - \frac{2}{x^2-1} \right]$$

4. Evaluate each of the following limits :

$$(a) \lim_{x \rightarrow 0} \frac{\sqrt{4+x} - \sqrt{4-x}}{x}$$

$$(b) \lim_{x \rightarrow 0} \frac{\sqrt{2+x} - \sqrt{2}}{x}$$

$$(c) \lim_{x \rightarrow 3} \frac{\sqrt{3+x} - \sqrt{6}}{x-3}$$

$$(d) \lim_{x \rightarrow 0} \frac{x}{\sqrt{1+x} - 1}$$

$$(e) \lim_{x \rightarrow 2} \frac{\sqrt{3x-2} - x}{2 - \sqrt{6-x}}$$

5. (a) Find $\lim_{x \rightarrow 0} \frac{2}{x}$, if it exists.

(b) Find $\lim_{x \rightarrow 2} \frac{1}{x-2}$, if it exists.

6. Find the values of the limits given below :

$$(a) \lim_{x \rightarrow 0} \frac{x}{5 - |x|}$$

$$(b) \lim_{x \rightarrow 2} \frac{1}{|x + 2|}$$

$$(c) \lim_{x \rightarrow 2} \frac{1}{|x - 2|}$$

$$(d) \text{ Show that } \lim_{x \rightarrow 5} \frac{|x - 5|}{x - 5} \text{ does not exist.}$$

7. (a) Find the left hand and right hand limits of the function

$$f(x) = \begin{cases} -2x + 3, & x \leq 1 \\ 3x - 5, & x > 1 \end{cases} \text{ as } x \rightarrow 1$$

$$(b) \text{ If } f(x) = \begin{cases} x^2, & x \leq 1 \\ 1, & x > 1 \end{cases}, \text{ find } \lim_{x \rightarrow 1} f(x)$$

$$(c) \text{ Find } \lim_{x \rightarrow 4} f(x) \text{ if it exists, given that } f(x) = \begin{cases} 4x + 3, & x < 4 \\ 3x + 7, & x \geq 4 \end{cases}$$

8. Find the value of 'a' such that $\lim_{x \rightarrow 2} f(x)$ exists, where $f(x) = \begin{cases} ax + 5, & x < 2 \\ x - 1, & x \geq 2 \end{cases}$

9. Let $f(x) = \begin{cases} x, & x < 1 \\ 1, & x = 1 \\ x^2, & x > 1 \end{cases}$

Establish the existence of $\lim_{x \rightarrow 1} f(x)$.

10. Find $\lim_{x \rightarrow 2} f(x)$ if it exists, where

$$f(x) = \begin{cases} x - 1, & x < 2 \\ 1, & x = 2 \\ x + 1, & x > 2 \end{cases}$$

Infinite limits and limit at infinity

Definition: Let f be defined on some open interval (a, c) .

- a. If for every number N there is some $\delta > 0$ such that if $0 < x - a < \delta$, then $f(x) > N$.

In that case we can write $\lim_{x \rightarrow a^+} f(x) = \infty$

- b. If for every number N , there is some $\delta > 0$ such that if $0 < x - a < \delta$, then $f(x) < N$

In that case we can write $\lim_{x \rightarrow a^+} f(x) = -\infty$

Definition: Let f be defined on some open interval (c, a) .

- a. If for every number N there is some $\delta < 0$ such that if $-\delta < x - a < 0$, then

$f(x) > N$. In this case we write $\lim_{x \rightarrow a^-} f(x) = \infty$.

- b. If for every number N there is some $\delta > 0$ such that if $-\delta < x - a < 0$, then

$f(x) < N$. In that case we write $\lim_{x \rightarrow a^-} f(x) = -\infty$.

Definition: (a) Let f be defined on an interval (a, ∞) . A number L is the limit of $f(x)$ as x approaches ∞ if for every $\varepsilon > 0$ there is a number M such that if $x > M$, then $|f(x) - L| < \varepsilon$. In this case we write $\lim_{x \rightarrow \infty} f(x) = L$. We say that the limit of $f(x)$ exists as x approaches ∞ , or that f has a limit at ∞ .

(b) Let f be defined on an interval $(-\infty, a)$. A number L is the limit of $f(x)$ as x approaches $-\infty$ if for every $\varepsilon > 0$ there is a number M such that if $x < M$, then $|f(x) - L| < \varepsilon$. In this case we write $\lim_{x \rightarrow -\infty} f(x) = L$. We say that the limit of $f(x)$ exists as x approaches $-\infty$ or that f has a limit at $-\infty$.

Example: Using $(\varepsilon - \delta)$ definitions prove the following limits

$$a. \lim_{x \rightarrow \frac{2}{3}^-} \frac{1}{3x-2} = -\infty \quad b. \lim_{x \rightarrow \frac{2}{3}^+} = \infty \quad c. \lim_{x \rightarrow \infty} \frac{2x+1}{x+2} = 2 \quad d. \lim_{x \rightarrow 5^-} \frac{6}{x-5} = -\infty$$

$$e. \lim_{x \rightarrow -\infty} \frac{x}{x-1} = 1 \quad f. \lim_{x \rightarrow 2^-} \frac{1}{(3x-6)^2} = \infty.$$

Solutions: a) Let $N < 0$ be given, we need to find $\delta > 0$ such that

$$-\delta < x - \frac{2}{3} < 0 \Rightarrow \frac{1}{3x-2} < N. \text{ But } \frac{1}{3x-2} < N \Rightarrow 3x-2 > \frac{1}{N} \Rightarrow x - \frac{2}{3} > \frac{1}{3N}$$

$$\text{Now, choose } \delta = -\frac{1}{3N}. \text{ Therefore } \lim_{x \rightarrow \frac{2}{3}^-} \frac{1}{3x-2} = -\infty.$$

c) Let $\varepsilon > 0$ be given, we need to find $M > 0$ such that

$$x > M \Rightarrow \left| \frac{2x+1}{x+2} - 2 \right| < \varepsilon. \text{ But } \left| \frac{2x+1}{x+2} - 2 \right| = \left| \frac{-3}{x+2} \right| = \frac{3}{x+2} < \frac{3}{x} < \varepsilon$$

$$\Rightarrow x > \frac{3}{\varepsilon}. \text{ Now, choose } M = \frac{3}{\varepsilon}. \text{ Therefore } \lim_{x \rightarrow \infty} \frac{2x+1}{x+2} = 2.$$

Continuity of a function

Definition: A function f is continuous at a point a in its domain if $\lim_{x \rightarrow a} f(x) = f(a)$

The definition implicitly requires three things if f is continuous at a ,

- i) $f(a)$ is defined
- ii) $\lim_{x \rightarrow a} f(x)$ exists
- iii) $\lim_{x \rightarrow a} f(x) = f(a)$

Definition: A function f is discontinuous at a point a in its domain if f is not continuous at a .

Example:

Determine whether the following function is continuous at $x=0$ or not.

$$f(x) = \begin{cases} \frac{x-6}{x-3} & \text{if } x < 0 \\ 2 & \text{if } x = 0 \\ \sqrt{4+x^2} & \text{if } x > 0 \end{cases}$$

Solution: f is defined at $x=0$ since $f(0)=2$. The left hand limit

$$\lim_{x \rightarrow 0^-} f(x) = \lim_{x \rightarrow 0^-} \frac{x-6}{x-3} = 2 \text{ and the right hand limit } \lim_{x \rightarrow 0^+} f(x) = \lim_{x \rightarrow 0^+} \sqrt{4+x^2} = 2.$$

Thus, $\lim_{x \rightarrow 0} f(x)$ exist with $\lim_{x \rightarrow 0} f(x) = 2$. Moreover, $\lim_{x \rightarrow 0} f(x) = 2 = f(0)$

All the three conditions are satisfied and f is continuous at $x=0$.

Examples: a) Find the values of a and b that make f continuous everywhere.

$$f(x) = \begin{cases} \frac{x^2 - 4}{x - 2} & \text{if } x < 2 \\ ax^2 - bx + 3 & \text{if } 2 \leq x < 3 \\ 2x - a + b & \text{if } x \geq 3 \end{cases}$$

b) For what values of the constant a is the function f continuous on $(-\infty, \infty)$?

$$f(x) = \begin{cases} ax^2 + 2x & \text{if } x < 2 \\ x^3 - ax & \text{if } x \geq 2 \end{cases}$$

c) Give a necessary and sufficient condition on C and D for a function

$$f(x) = \begin{cases} Cx + D & \text{for } x < -1 \\ 2x & \text{for } -1 \leq x \leq 2 \\ 2Dx - C & \text{for } 2 < x \end{cases} \quad \text{to be continuous at } x = -1 \text{ and not continuous at } x = 2.$$

Definition: A function f is continuous from the right at a point a in its domain if

$\lim_{x \rightarrow a^+} f(x) = f(a)$. And a function f is continuous from the left at a point a in its domain if

$\lim_{x \rightarrow a^-} f(x) = f(a)$.

Example:

Show that the diving board function, defined by $f(x) = \begin{cases} 0 & \text{for } x < 0 \\ 1 & \text{for } x \geq 0 \end{cases}$

is continuous from the right at 0 but not continuous from the left at 0.

Solution: Observe that f is defined at 0 since $f(0)=1$. The right hand limit

$\lim_{x \rightarrow 0^+} f(x) = \lim_{x \rightarrow 0^+} 1 = 1 = f(0)$. Thus f is continuous from the right at 0.

But the left hand limit $\lim_{x \rightarrow 0^-} f(x) = \lim_{x \rightarrow 0^-} 0 = 0 \neq 1 = f(0)$ which implies that f is not

continuous from the left at 0.

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Definition: A function is continuous on an open interval (a,b) if it is continuous at every point in (a,b) and it is continuous on a closed interval $[a,b]$ if it is continuous at every point in (a,b) and is also continuous from the right at a and continuous from the left at b .

Example: Show that $f(x) = \sqrt{1-x^2}$ is continuous on $[-1,1]$

Solution: Since $\sqrt{x}$ is defined for $x \geq 0$ and since $1-x^2 \geq 0$ if and only if $-1 \leq x \leq 1$, it follows that the domain of f is $[-1, 1]$.

Now, $\lim_{x \rightarrow -1^+} f(x) = \lim_{x \rightarrow -1^+} \sqrt{1-x^2} = 0 = f(-1)$ and $\lim_{x \rightarrow 1^-} f(x) = \lim_{x \rightarrow 1^-} \sqrt{1-x^2} = 0 = f(1)$

Next we observe that for every a in $(-1, 1)$ $\lim_{x \rightarrow a} \sqrt{1-x^2} = \sqrt{1-a^2} = f(a)$ which implies

that f is continuous at every point in $(-1, 1)$. Hence f is continuous on a closed interval $[-1,1]$.

Theorem (Intermediate Value theorem)

Suppose f is continuous on a closed interval $[a, b]$. Let P be any number between

$f(a) \leq P \leq f(b)$ or $f(b) \leq P \leq f(a)$. Then there exists a number c in $[a, b]$ such that $f(c) = P$

Example:

Use the Intermediate Value Theorem to prove the following statements

i) $x^3 - 5x - 17 = 0$ has a root between 3 and 4

ii) The function $f(x) = \frac{x^6}{x + 43}$ takes the value 1 for some x in $[1, 2]$

iii) $\sqrt{x} + \sqrt{x+1} = 5$ for some real number x in $[1, 8]$

Example:

Show that there is a root of the equation $4x^3 - 6x^2 + 3x - 2 = 0$ between 1 and 2

Solution: Let $f(x) = 4x^3 - 6x^2 + 3x - 2$. We are looking for a solution of the given equation that is a number c between 1 and 2 such that $f(c) = 0$ therefore, we take $a=1$, $b=2$ and $N=0$ in the above theorem. We have $f(1) = 4 - 6 + 3 - 2 = -1 < 0$ and $f(2) = 32 - 24 + 6 - 2 = 12 > 0$.

Thus $f(1) < 0 < f(2)$, that is $N=0$ is a number between $f(1)$ and $f(2)$. Now f is continuous since it is a polynomial, so the IVT says that there is a number c between 1 and 2 such that $f(c)=0$.

Exercises: 1) Use the IVT to show that there is a root of the given equation in the specified interval.

a. $\cos x = x$, $(0, 1)$ b. $\ln x = e^{-x}$, $(1, 2)$